Chapter 3.5

Extensions of the Cox proportional hazards model

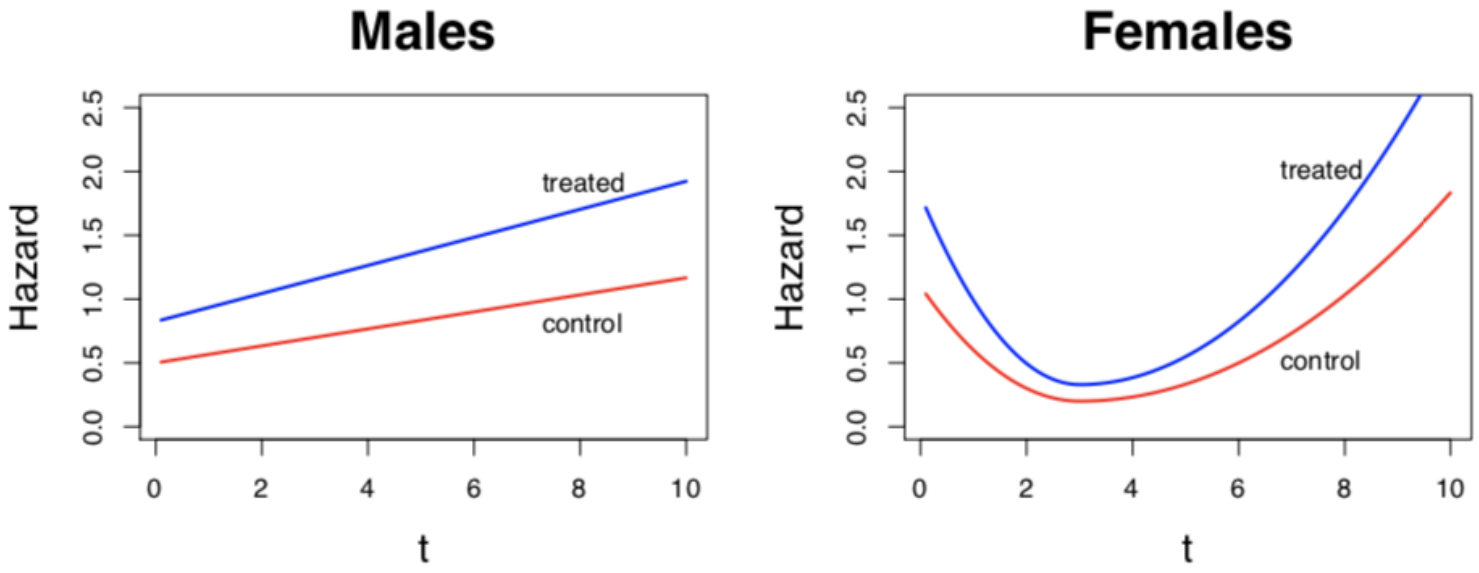
Biostats 2

What if the proportional hazards assumption is violated?

- Partition time into intervals
- Stratify
- Use time-dependent covariates

Stratified Cox Regression

- HR for treatment vs placebo is constant for females and males, respectively.
- HR for female vs male may not be constant.
- The shapes of baseline hazard functions across strata may be different.



Stratified Cox regression

- A useful extension of the standard Cox model when a covariate does not satisfy the proportional hazards assumption
- It allows us to adjust for a factor without estimating its effect directly, that is, without reporting a coefficient or a p-value for that factor
- The proportional hazards assumption may fail overall, but still be reasonable within each stratum
- This is a semi-parametric model with a separate nonparametric baseline hazard function in each stratum
- The model estimates a common $\hat{\boldsymbol{\beta}}$ across strata

Stratified Cox regression, continued

Suppose there is a factor with K strata. In a stratified Cox model, the hazard for stratum k is

$$h_k(t) = h_{0k}(t) \exp(\mathbf{x}^\top \boldsymbol{\beta})$$

where $h_{0k}(t)$ is the baseline hazard for stratum k, $k = 1, \ldots, K$.

Key assumption

- Covariate effects are the same across strata
- If this is not true, that is, if there is an interaction between the stratification variable and the predictors, then stratum-specific estimates should be reported. In that case, fitting separate Cox models may be more appropriate.

- The proportional hazards assumption can be checked within each stratum
- Estimation combines the partial likelihood contributions across strata and then finds the MLE of $\boldsymbol{\beta}$
- Interpretation of β: log hazard ratio within a stratum

Example

- Colon cancer stage II and stage III patients may have different prognoses
- We may also want to combine results across multiple study centers

Why not run two Cox models?

- Interpretation: we may not want the treatment effect to vary by study center. Also, what would we do for a new hospital?
- Power: stratified Cox regression pools information across strata

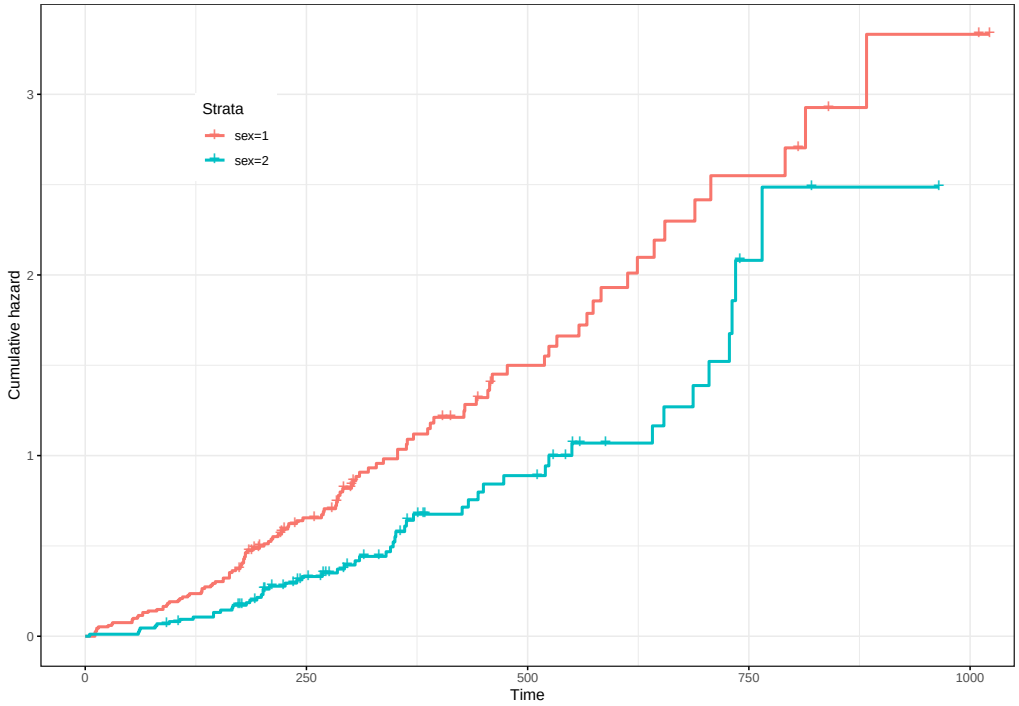
Comments on the stratified Cox model

- There is no direct way to carry out inference for the stratification variable itself
- If a confounder is controlled through stratification, there is no hazard ratio estimate comparing levels of that confounder
- This becomes cumbersome with continuous variables and with multiple stratification variables, because there is no simple additive structure for the baseline hazards. For example, with two stratification variables having n_1 and n_2 categories, there are $n_1 \times n_2$ strata in total.
- As the number of strata increases, the number of observations per stratum decreases, making the baseline hazard harder to estimate for prediction
- Unnecessary stratification, even when the proportional hazards assumption holds, reduces estimation efficiency, although the loss is often small

Example

```r
head(lung) # NCCTG Lung Cancer Data
```

```
##   inst time status age sex ph.ecog ph.karno pat.karno meal.cal wt.loss
## 1    3  306      2  74   1       1       90       100     1175      NA
## 2    3  455      2  68   1       0       90        90     1225      15
## 3    3 1010      1  56   1       0       90        90       NA      15
## 4    5  210      2  57   1       1       90        60     1150      11
## 5    1  883      2  60   1       0      100        90       NA       0
## 6   12 1022      1  74   1       1       50        80      513       0
```



The cumulative hazard functions by gender appear to cross, suggesting possible nonproportional hazards.

```r
cox_s = coxph(Surv(time, status) ~ age + wt.loss + strata(sex), data = lung)
summary(cox_s)
```

```
## Call:
## coxph(formula = Surv(time, status) ~ age + wt.loss + strata(sex),
##     data = lung)
##
##   n= 214, number of events= 152
##    (14 observations deleted due to missingness)
##
##              coef exp(coef)  se(coef)     z Pr(>|z|)
## age     0.0192190 1.0194049 0.0096226 1.997   0.0458 *
## wt.loss 0.0001412 1.0001412 0.0062509 0.023   0.9820
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##         exp(coef) exp(-coef) lower .95 upper .95
## age         1.019     0.9810     1.000     1.039
## wt.loss     1.000     0.9999     0.988     1.012
##
## Concordance= 0.561  (se = 0.027 )
## Likelihood ratio test= 4.09  on 2 df,   p=0.1
## Wald test            = 3.99  on 2 df,   p=0.1
## Score (logrank) test = 4  on 2 df,   p=0.1
```

- One common set of $\hat{\boldsymbol{\beta}}$
- No coefficient estimate for sex

Time-dependent covariates

Recall the Cox model with time-independent covariates:

$$h(t) = h_0(t)e^{\beta_1 x_1 + \cdots + \beta_p x_p}$$

- The baseline hazard $h_0(t)$ depends on time t, but not on the covariates $(x_1, \ldots, x_p)$
- The hazard ratio depends on the covariates $(x_1, \ldots, x_p)$, but not on time t

What if some covariates are measured over time, so that their values can change?

Examples of time-dependent covariates

- Internal covariates
  - Variables related to the individual
  - Measured only while the individual is still alive and under follow-up
  - Examples: white blood cell count, blood pressure, disease complications
- External (ancillary) covariates
  - Variables that change in a known way, for example age
  - Variables that exist independently of the individual, for example temperature or air pollution

With time-dependent covariates $(z_1(t), \ldots, z_q(t))$, we write the Cox model as

$$h(t) = h_0(t)e^{\beta_1 z_1(t) + \cdots + \beta_q z_q(t)}.$$

- The hazard at time t depends on the values of the covariates at that time
- The regression coefficient for covariate $z_j(t)$ is constant over time
- Because the covariates change over time, the hazard ratio can also vary over time
- In this course, we will not consider models with both time-dependent covariates and time-dependent coefficients at the same time

- Time-varying covariates must be constructed carefully to ensure meaningful interpretation
- If a covariate changes deterministically with study time, there may be little value in adding it as a time-varying covariate by itself

Model interpretation

Time-independent	Compare the risk of the event between subjects defined by a fixed exposure status, usually based on baseline or overall exposure classification.
Time-dependent	Compare the risk of the event at each event time, updating each subject's exposure status according to whether that subject has been exposed by that time.

Example: Heart Transplantation

A group of patients eligible for heart transplantation is recruited. However, a heart must first become available, and then the patient with the closest match receives the transplant.

The scientific question is:

How should we evaluate the effect of heart transplantation on survival?

A key complication is that transplant status is not determined at baseline. A patient may receive a transplant only after surviving long enough for a donor heart to become available.

Naive Comparison and Immortal Time Bias

Suppose we compare, from enrollment onward,

- patients who eventually received a transplant, versus
- patients who never received a transplant.

This creates **immortal time bias**.

A patient classified as eventually transplanted must survive until the transplant time. Therefore, the period between enrollment and transplant is an **immortal time** for that group: if a patient dies during that period, that patient can no longer belong to the transplanted group.

As a result, the transplanted group can appear to have better survival simply because its members were required to survive long enough to receive treatment.

An Incorrect Definition of Survival Time

We could define survival time as follows:

- Time from transplant to death for those who received a transplant
- Time from enrollment to death for those who did not receive a transplant

But then the two groups are not comparable.

This definition favors the transplant group because the waiting time before transplant is excluded from their survival time. Patients in the transplant group had to survive that waiting period, but that time is not counted in the analysis.

Time-Varying Covariate

A principled solution is to treat transplant status as a **time-varying covariate**:

$$Z(t) = \begin{cases} 1, & \text{if a patient has received a transplant by time } t, \\ 0, & \text{otherwise.} \end{cases}$$

Then fit a time-varying Cox model:

$$h(t) = h_0(t) \exp\{\beta Z(t)\}.$$

Interpretation:

- before transplant, a patient contributes risk time to the $Z(t) = 0$ group
- after transplant, the same patient contributes risk time to the $Z(t) = 1$ group

This correctly aligns treatment status with follow-up time.

Example 2

```r
head(aids) # Didanosine versus Zalcitabine in HIV Patients
```

```
##   patient  Time death       CD4 gender prevOI start  stop event
## 1       1 16.97     0 10.677078   male   AIDS     0  6.00     0
## 2       1 16.97     0  8.426150   male   AIDS     6 12.00     0
## 3       1 16.97     0  9.433981   male   AIDS    12 16.97     0
## 4       2 19.00     0  6.324555   male noAIDS     0  6.00     0
## 5       2 19.00     0  8.124038   male noAIDS     6 12.00     0
## 6       2 19.00     0  4.582576   male noAIDS    12 18.00     0
```

```r
head(aids2) # Data for time-independent Cox model
```

```
##   patient  Time death      CD4 gender prevOI event
## 1       1 16.97     0 9.433981   male   AIDS     0
## 2       2 19.00     0 5.000000   male noAIDS     0
## 3       3 18.53     1 6.164414 female   AIDS     1
## 4       4 12.70     0 1.732051   male   AIDS     0
## 5       5 15.13     0 6.708204   male   AIDS     0
## 6       6  1.90     1 4.582576 female   AIDS     1
```

```r
# Time-independent
cox_indep = coxph(Surv(Time, death) ~ CD4, data = aids2)
summary(cox_indep)
```

```
## Call:
## coxph(formula = Surv(Time, death) ~ CD4, data = aids2)
##
##   n= 467, number of events= 188
##
##         coef exp(coef) se(coef)      z Pr(>|z|)
## CD4 -0.14395   0.86593  0.02302 -6.252 4.05e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##     exp(coef) exp(-coef) lower .95 upper .95
## CD4    0.8659      1.155    0.8277    0.9059
##
## Concordance= 0.631  (se = 0.021 )
## Likelihood ratio test= 53.46  on 1 df,   p=3e-13
## Wald test            = 39.09  on 1 df,   p=4e-10
## Score (logrank) test = 42.23  on 1 df,   p=8e-11
```

```r
# Time-dependent
cox_dep = coxph(Surv(start, stop, event) ~ CD4, data = aids)
summary(cox_dep)
```

```
## Call:
## coxph(formula = Surv(start, stop, event) ~ CD4, data = aids)
##
##   n= 1405, number of events= 188
##
##         coef exp(coef) se(coef)      z Pr(>|z|)
## CD4 -0.19040   0.82663  0.02423 -7.858 3.92e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##     exp(coef) exp(-coef) lower .95 upper .95
## CD4    0.8266       1.21    0.7883    0.8668
##
## Concordance= 0.691  (se = 0.018 )
## Likelihood ratio test= 90.15  on 1 df,   p=<2e-16
## Wald test            = 61.74  on 1 df,   p=4e-15
## Score (logrank) test = 69.04  on 1 df,   p=<2e-16
```

Example: Landmark Analysis in Lung Cancer

A study enrolls patients with advanced lung cancer at the start of immunotherapy.

Tumor response is evaluated at the first scheduled imaging assessment, 3 months after treatment initiation.

At that time, patients are classified as

- **Responder at 3 months**
- **Nonresponder at 3 months**

The scientific question is:

Among patients who survive to the 3-month scan, does subsequent survival differ by response status at 3 months?

Why a Naive Analysis is Problematic

Suppose we compare, from treatment initiation onward,

- patients who are eventually classified as responders at 3 months, versus
- patients who are not.

This is problematic because a patient must survive long enough to reach the 3-month imaging assessment in order to be classified as a responder.

Therefore, patients in the responder group are guaranteed to survive to that assessment time, creating a form of **immortal time bias**.

So a naive comparison from time 0 would unfairly favor the responder group.

Landmark Analysis

A simple way to address this issue is **landmark analysis**.

Choose a fixed landmark time

$$t_L = 3 \text{ months}.$$

Then:

- include only patients who are alive and under follow-up at t_L
- classify each patient by tumor response status at t_L
- compare survival from t_L onward

In this way, all patients in the comparison have survived to the same time point before group membership is defined.

How the Landmark Analysis is Performed

At the 3-month landmark:

- exclude patients who died or were lost to follow-up before 3 months
- define
  - $Z = 1$: responder at 3 months
  - $Z = 0$: nonresponder at 3 months
- reset time origin to the landmark time

Then compare post-landmark survival using methods such as:

- Kaplan–Meier curves starting at 3 months
- log-rank test
- Cox regression with response status at 3 months as a fixed covariate

Interpretation of Landmark Analysis

Landmark analysis answers a **conditional prognosis** question:

Among patients alive at 3 months, how does future survival differ by response status at 3 months?

It does **not** answer:

- the effect of immunotherapy itself
- survival differences from treatment initiation in the full study population

So the interpretation is conditional on having survived to the landmark time.

Strengths and Limitations

Strengths

- Easy to understand and implement
- Avoids immortal time bias
- Matches a clinically meaningful question at a fixed visit

Limitations

- Excludes patients who fail before the landmark time
- May reduce statistical efficiency
- Results depend on the choice of landmark time

Take-Home Message

- In this lung cancer example, tumor response is measured at a fixed post-baseline time point
- A naive comparison from time 0 is biased because responders must survive long enough to be assessed
- Landmark analysis avoids this problem by comparing patients only after a common landmark time
- The resulting comparison is a prognosis comparison among patients alive at 3 months

Time-Varying Covariate vs Landmark Analysis

Feature	Time-varying covariate	Landmark analysis
Treatment status	Can change over time	Fixed at a chosen landmark time
Uses full follow-up?	Yes	No, excludes those failing before t_L
Avoids immortal time bias?	Yes, if coded correctly	Yes, by design
Interpretation	Effect of current treatment status on hazard	Survival comparison among those alive at t_L
Efficiency		Simpler but may lose information
Complexity	More technical	Easier to explain and present

Which Method Answers Which Question?

Time-varying covariate:

At time t, how does the hazard differ between patients who have already received a transplant and those who have not?

Landmark analysis:

Among patients who survive to time t_L, how does future survival differ by response status at t_L?

So the two approaches are related, but they do not answer exactly the same scientific question.

Take-Home Messages

- **Immortal time bias** arises when treatment assignment uses future information and incorrectly credits guaranteed survival time to the treated group.
- A **time-varying covariate** is often the most principled way to analyze treatments that begin during follow-up.
- **Landmark analysis** is a method that compares patients from a common time point onward.
- Time-varying Cox models usually use data more efficiently, whereas landmark analysis is easier to explain and visualize.

What is a nested case-control study?

- A nested case-control study is a case-control sample drawn from a well-defined cohort.
- The cohort is assembled first and followed over time for the outcome.
- Whenever a case occurs, controls are sampled from the **risk set**, meaning cohort members who are still at risk at that time.
- It is especially useful when the exposure is expensive or difficult to measure, for example, biomarkers, genetic assays, or chart review variables.

Key idea: we keep the cohort structure and event times, but only measure detailed exposures on all cases and a sample of eligible controls.

Why use this design?

- Full cohort analysis may be expensive if exposure data must be collected retrospectively or assayed in stored specimens.
- A nested case-control design greatly reduces cost and effort.
- Because controls come from the same cohort, selection bias is often less severe than in a traditional case-control study.
- Temporality is preserved: exposure information is defined prior to disease occurrence.

Common use cases

- Stored blood samples in a prospective cohort
- Expensive molecular profiling
- Manual abstraction of detailed clinical covariates

How the sampling works

1. Start with a cohort free of the outcome at baseline.
2. Follow subjects over time.
3. Each time a case occurs at time t_j, define the **risk set** $R(t_j) = \{i :$ subject i is still under observation and outcome-free just before $t_j\}$.
4. Select the case and sample one or more controls from $R(t_j)$.
5. Repeat this for every case time.

Important details

- A person can serve as a control more than once.
- A person selected earlier as a control can later become a case.
- Matching is often done on follow-up time and sometimes on age, sex, or calendar time.

Simple example

Suppose a cohort study follows 20,000 participants for incident cardiovascular disease.

- Measuring an inflammatory biomarker on everyone is too expensive.
- During follow-up, each time a participant develops cardiovascular disease, we:
  - include that participant as a case,
  - sample 2 or 3 controls from participants still disease-free at that same time.
- The biomarker is then assayed only for:
  - all sampled cases,
  - sampled controls.

This yields a much smaller analytic dataset while retaining the time-matched sampling structure from the full cohort.

What parameter does it estimate?

With risk-set sampling, nested case-control analysis targets the same regression parameter as a Cox proportional hazards model fit to the full cohort.

If

$$h(t \mid X) = h_0(t) \exp(\beta X),$$

then conditional logistic regression on matched case-control sets estimates β.

Interpretation

- $\exp(\beta)$ is interpreted as a hazard ratio.
- This is not merely a rare-disease approximation.
- The equivalence comes from sampling controls from the risk set at each event time.

Analysis

At each failure time t_j, form a matched risk set consisting of

- the case who fails at time t_j, and
- m controls sampled from those still at risk at time t_j.

Let $Y_{ij} = 1$ if subject i is the case in matched set j, and $Y_{ij} = 0$ otherwise. A stratified logistic regression model is

$$\text{logit}\left\{\Pr(Y_{ij} = 1 \mid X_{ij}, j)\right\} = \alpha_j + \beta^\top X_{ij},$$

where α_j is a matched-set-specific intercept.

Conditioning on there being exactly one case in set j gives

$$\Pr(\text{case is subject } i \mid \text{matched set } j) = \frac{\exp(\beta^\top X_{ij})}{\sum_{\ell \in \text{set } j} \exp(\beta^\top X_{\ell j})}.$$

Why conditional logistic regression?

- It compares subjects only within the same matched risk set.
- It removes the nuisance intercept α_j.
- It naturally accounts for matching on failure time.
- Its likelihood has the same form as the Cox partial likelihood.

Quiz

- Show that, in a nested case-control study, increasing the number of sampled controls per case does not materially change the estimate of the regression coefficient.
- Show that increasing the number of sampled controls per case reduces the width of the confidence interval.
- Explain why this gain in precision becomes smaller as the number of sampled controls continues to increase.

Advantages and limitations

Advantages

- Cost efficient
- Temporality preserved
- Less selection bias than standard case-control studies
- Efficient for rare outcomes or expensive exposures
- Can estimate hazard ratio parameters

Limitations

- More complex sampling and analysis
- Some loss of efficiency compared with full cohort analysis
- Absolute risks are not directly estimated from the sampled data alone
- Repeated control sampling must be handled correctly

Take-home points

- A nested case-control study is a cost-efficient subsampling design within a cohort.
- Controls are sampled from the risk set when each case occurs.
- A subject may be sampled as a control multiple times and may later become a case.
- Conditional logistic regression is the standard analysis for matched sets.
- Under risk-set sampling, the estimated association corresponds to the hazard ratio from the full cohort Cox model.

HR and OR

Let group $g \in \{0, 1\}$ denote control and treatment.

Hazard ratio

$$\mathrm{HR}(t) = \frac{h_1(t)}{h_0(t)}$$

is a ratio of **instantaneous event rates**.

Odds ratio at a fixed horizon t

$$\mathrm{OR}(t) = \frac{\dfrac{P(T \leq t \mid g = 1)}{1 - P(T \leq t \mid g = 1)}}{\dfrac{P(T \leq t \mid g = 0)}{1 - P(T \leq t \mid g = 0)}}$$

is a ratio of **cumulative event odds by time** t.

So HR concerns the *local rate*, whereas OR concerns the *accumulated probability* up to a time point.

Rare-event approximation: odds $\approx$ risk

Let

$$p_g(t) = P(T \leq t \mid g).$$

Then the event odds in group g is

$$\frac{p_g(t)}{1 - p_g(t)}.$$

If the event is rare by time t, then $p_g(t) \ll 1$, so

$$1 - p_g(t) \approx 1 \quad \Longrightarrow \quad \frac{p_g(t)}{1 - p_g(t)} \approx p_g(t).$$

Hence

$$\mathrm{OR}(t) = \frac{p_1(t)/(1 - p_1(t))}{p_0(t)/(1 - p_0(t))} \approx \frac{p_1(t)}{p_0(t)} = \mathrm{RR}(t).$$

So for rare events, **OR is close to the risk ratio (RR)**.

Rare-event approximation: risk $\approx$ cumulative hazard

Let

$$H_g(t) = \int_0^t h_g(u)\,du$$

be the cumulative hazard in group g. Then

$$S_g(t) = \exp\{-H_g(t)\}, \qquad p_g(t) = 1 - S_g(t) = 1 - e^{-H_g(t)}.$$

If the event is rare by time t, then $H_g(t)$ is small, and using

$$e^{-x} \approx 1 - x \quad \text{for small } x,$$

we get

$$p_g(t) = 1 - e^{-H_g(t)} \approx H_g(t).$$

Therefore,

$$\mathrm{RR}(t) = \frac{p_1(t)}{p_0(t)} \approx \frac{H_1(t)}{H_0(t)}.$$

From cumulative hazard ratio to HR

Suppose the proportional hazards model holds:

$$h_1(t) = \theta h_0(t),$$

where θ is constant. Then

$$H_1(t) = \int_0^t h_1(u)\,du = \int_0^t \theta h_0(u)\,du = \theta H_0(t).$$

So

$$\frac{H_1(t)}{H_0(t)} = \theta = \mathrm{HR}.$$

Combining the earlier approximations gives

$$\mathrm{OR}(t) \approx \mathrm{RR}(t) \approx \frac{H_1(t)}{H_0(t)} = \mathrm{HR}.$$

Conclusion: when events are rare up to time t and hazards are approximately proportional, the fixed-time OR is close to the HR.

Simple constant-hazard derivation

Assume constant hazards over the interval:

$$h_g(t) = \lambda_g.$$

Then

$$p_g(t) = 1 - e^{-\lambda_g t} \approx \lambda_g t \qquad \text{when } \lambda_g t \text{ is small.}$$

Thus the odds is approximately

$$\frac{p_g(t)}{1 - p_g(t)} \approx p_g(t) \approx \lambda_g t.$$

Hence

$$\mathrm{OR}(t) \approx \frac{\lambda_1 t}{\lambda_0 t} = \frac{\lambda_1}{\lambda_0} = \mathrm{HR}.$$

This is the cleanest intuition: with low event probability, cumulative incidence is roughly hazard $\times$ time.

Numerical illustration

Suppose by one year:

$$p_0 = 0.02, \qquad p_1 = 0.01.$$

Then

$$\mathrm{RR} = \frac{0.01}{0.02} = 0.50.$$

The odds ratio is

$$\mathrm{OR} = \frac{0.01/0.99}{0.02/0.98} \approx 0.495.$$

These are extremely close because both risks are small.

Under constant hazards,

$$\lambda_0 \approx -\log(0.98) = 0.0202, \qquad \lambda_1 \approx -\log(0.99) = 0.0101,$$

so

$$\mathrm{HR} = \frac{\lambda_1}{\lambda_0} \approx 0.50.$$

Thus in this rare-event setting,

$$\mathrm{OR} \approx \mathrm{RR} \approx \mathrm{HR}.$$

When the approximation breaks down

The statement $\mathrm{HR} \approx \mathrm{OR}$ can fail when:

- the event is **not rare**, so odds is no longer close to risk;
- follow-up is long enough that cumulative incidence is not small;
- hazards are **not proportional** over time;
- censoring patterns or differing follow-up make fixed-time risks hard to compare directly.

In particular, OR can exaggerate the magnitude of association relative to RR when the event is common.

So it is better to say:

For rare events over a fixed time horizon, OR may approximate HR,

not that they are interchangeable in general.

Take-home summary

1. HR is a ratio of instantaneous hazards.
2. OR at time t is a ratio of cumulative event odds.
3. If events are rare, then

$$\text{odds} \approx \text{risk} \approx \text{cumulative hazard}.$$

4. Under proportional hazards,

$$\mathrm{OR}(t) \approx \mathrm{RR}(t) \approx \mathrm{HR}.$$

Rare event + proportional hazards + fixed horizon
$\Longrightarrow$ OR is often numerically close to HR.